Startup Success Prediction

Using Machine Learning to predict startup outcomes based on funding, team strength, market demand, and founder experience — transforming raw data into actionable investment intelligence.

AI-POWERED ANALYTICS

BINARY CLASSIFICATION

What This Project Does

📊 Data-Driven

Analyzes historical startup records to surface patterns linked to success or failure.

🤖 ML-Powered

Uses Machine Learning to evaluate multiple business factors simultaneously and rapidly.

🎯 Predictive

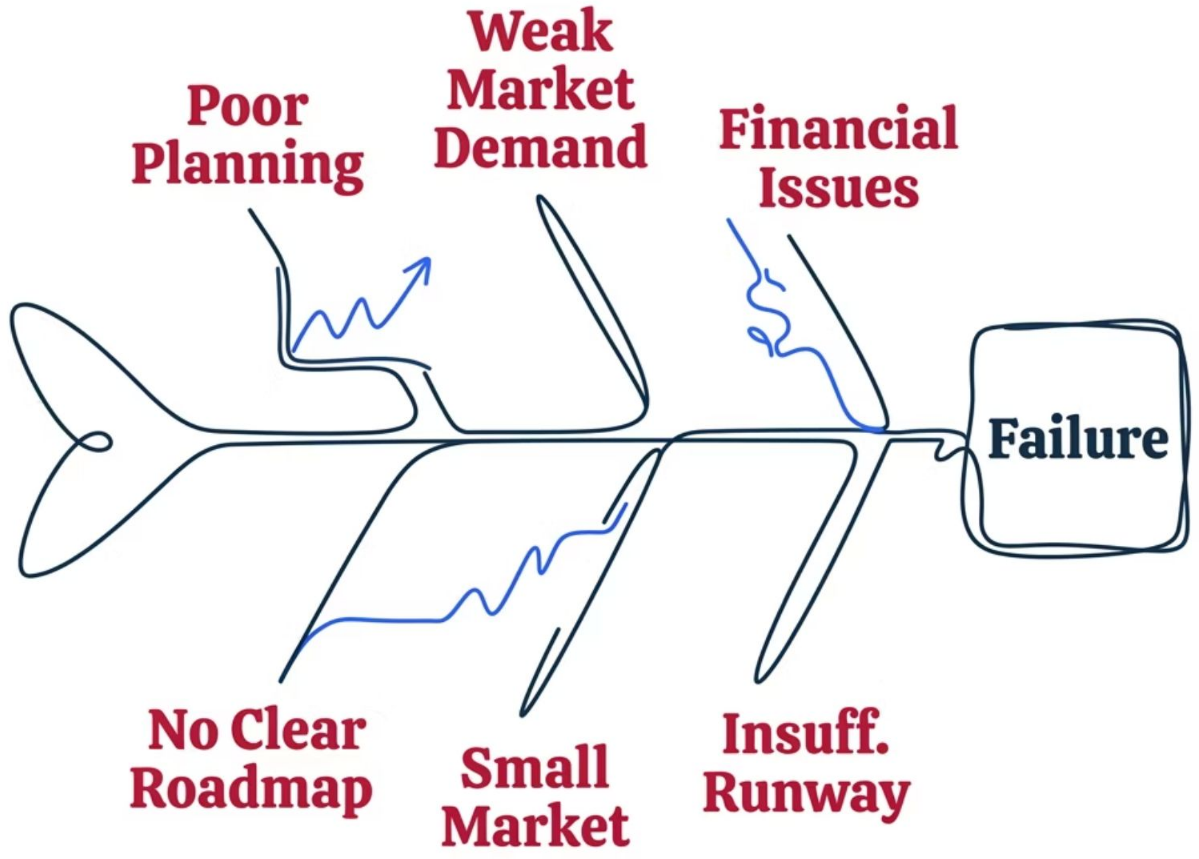
Outputs a success probability score based on funding, team size, revenue, and founder experience.

The Problem We're Solving

Why Startups Fail

Over **90% of startups** fail — yet most failures are predictable. Poor planning, weak market fit, and financial mismanagement are leading causes, but no scalable evaluation tool exists to catch these signals early.

Investors and founders lack a systematic, multi-factor tool to objectively assess startup potential before committing capital.



Defining Success: The Target

Label Mapping Logic

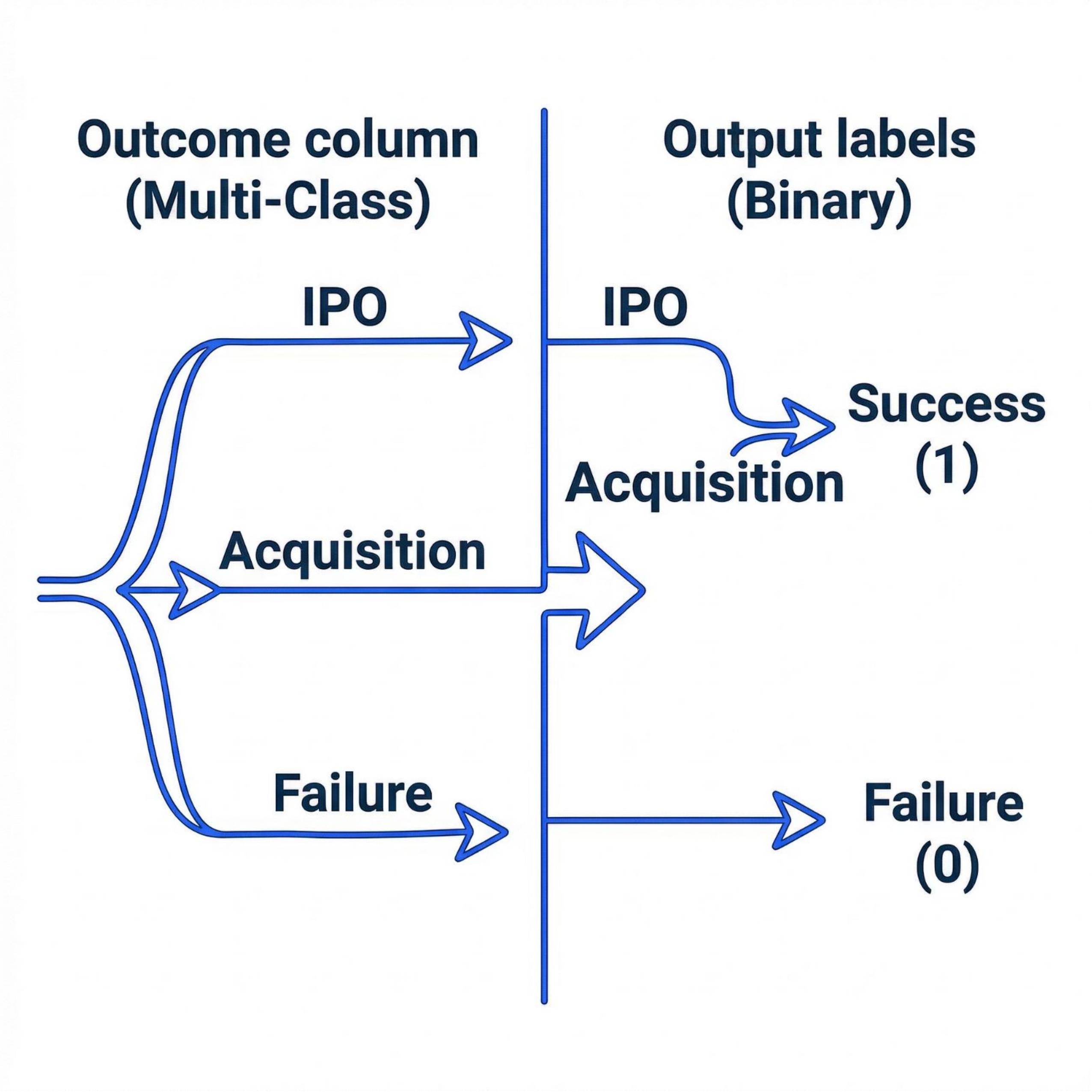
The **Outcome** column is converted into a binary classification label for model training.

IPO → **Success (1)**

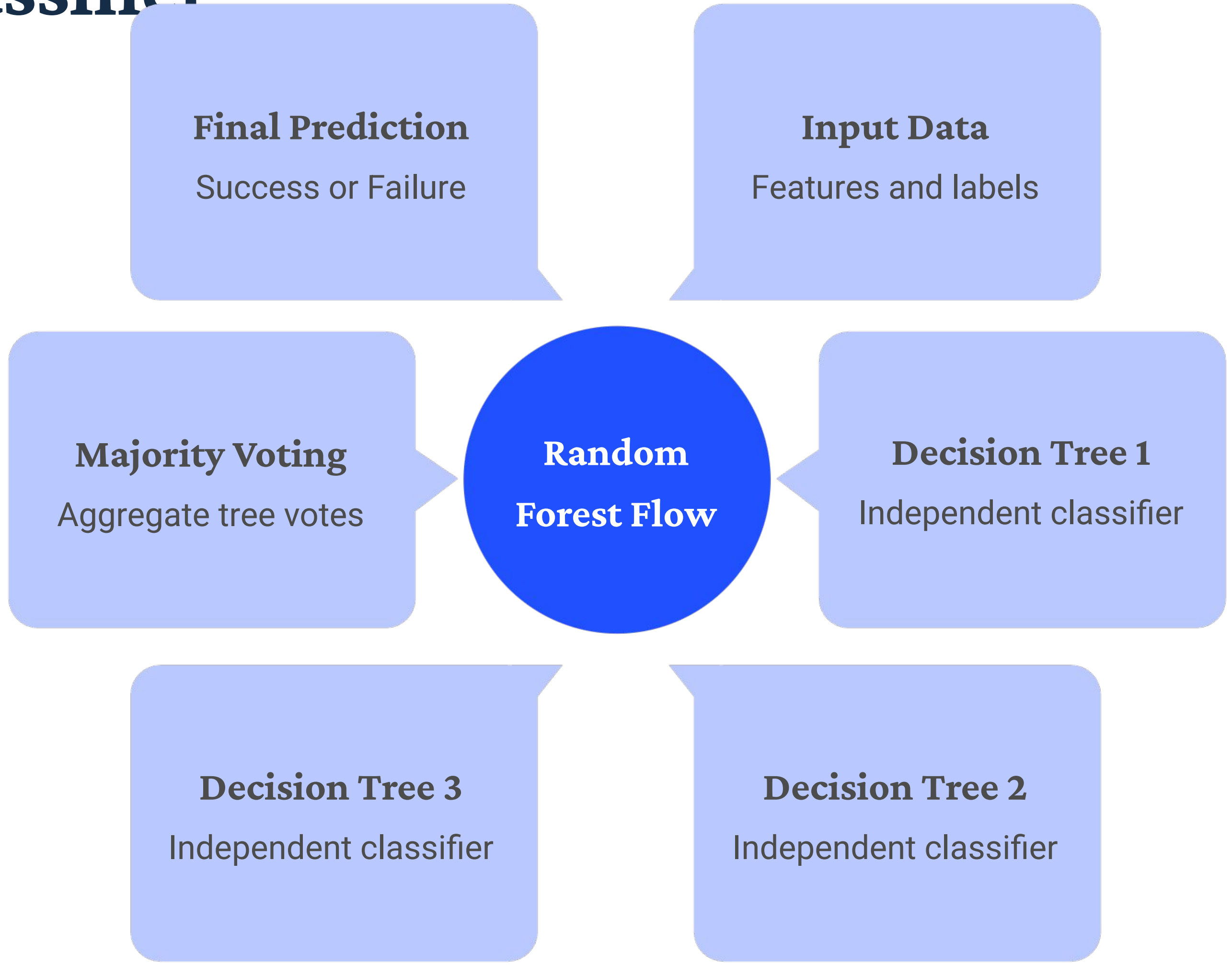
Acquisition → **Success (1)**

Failure → **Failure (0)**

This creates a clean **binary classification** problem the model can learn from.



Algorithm: Random Forest Classifier



Each decision tree votes independently; the majority output becomes the final classification — reducing overfitting and boosting reliability.

Why Random Forest?

High Accuracy

Ensemble voting outperforms single models on structured data.

Handles Scale

Efficient with large, high-dimensional datasets.

Feature Importance

Ranks which inputs drive predictions most.

Reduces Overfitting

Averaging multiple trees prevents memorization.

Model Performance & Feature

Evaluation Results
Importance

82%

Accuracy

Overall correct predictions on test set

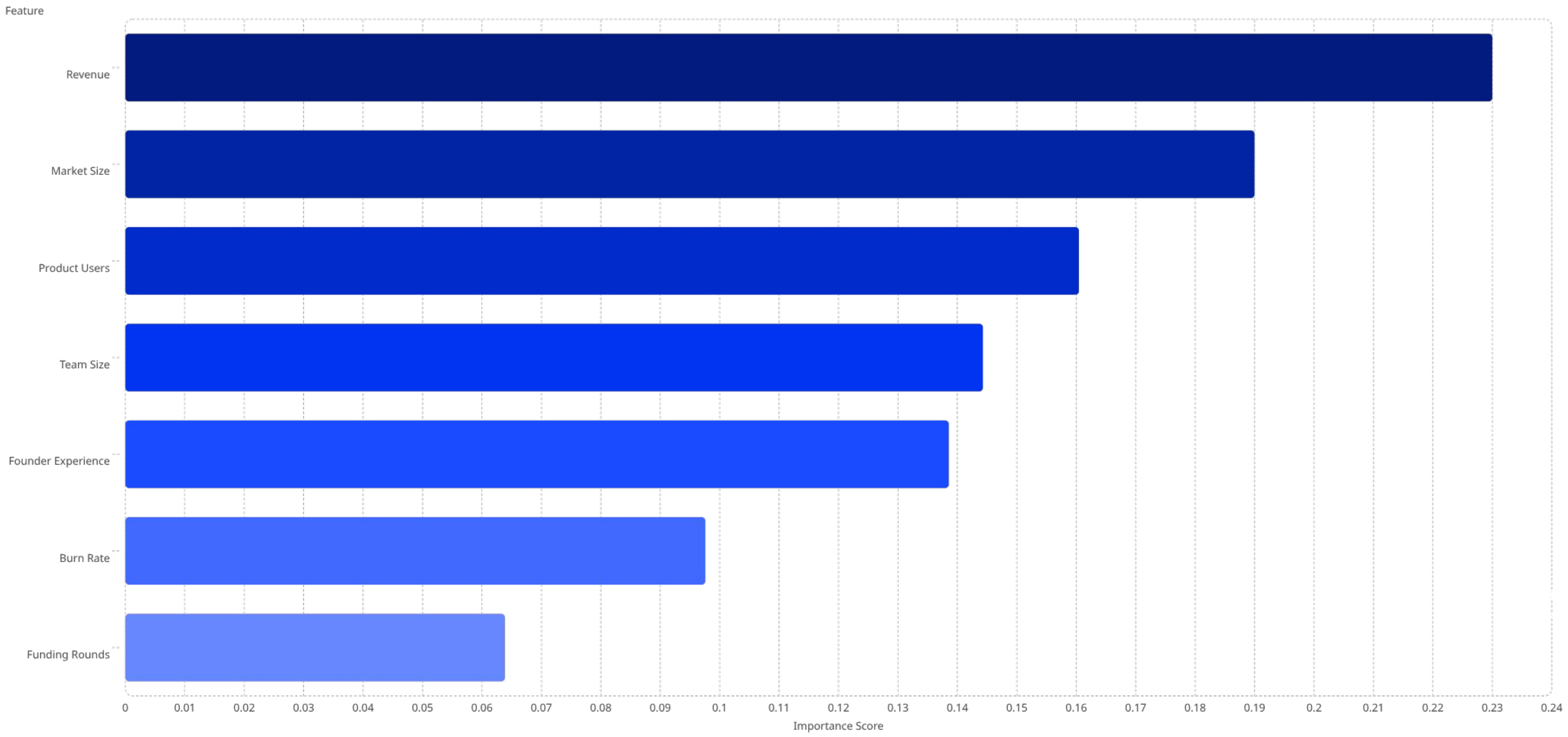
0.84

F1-Score

Balanced precision-recall performance

Confusion matrix and classification report confirm strong performance across both Success and Failure classes, with minimal false-negative risk on investment decisions.

Top Predictive Features



0.0
9

Project Team

Rajesh

Kishore

Moses